

Constrained Decoding Prevents Cardinality Degradation in Structured Extraction

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Abstract

Schema-constrained decoding is widely believed to cost accuracy, because forcing output through a grammar masks tokens the model would otherwise emit. We measure that cost along an axis it has not been measured on: the number of items the model must produce. Holding input length constant by construction, we build documents containing exactly k verbatim-validated sentences, each naming one entity, and vary k from 1 to 16. Extraction recall declines in k under unconstrained decoding, by 0.214 from $k \leq 3$ to $k \geq 16$ (Spearman -0.293 , $p = 0.011$), and does not decline at all under a JSON schema (-0.114 , i.e. recall rises; $p = 0.73$). Paired on the same 75 documents per arm, unconstrained decoding leads by 0.200 at $k = 1$ and trails by 0.128 at $k = 16$ ($p = 0.005$). The mechanism is visible in emission counts: at $k = 16$ the constrained arm emits 9.4 of 16 available items and the unconstrained arm 7.2, at equal precision. Free to choose when to stop, the model stops early, and stops earlier as the list lengthens; a schema removes that choice because a partially filled array does not parse. We also show that item count, item density and document length cannot be held fixed simultaneously, which determines whether any measurement on this axis is an upper or a lower bound.

1 Introduction

Ask a language model for a list and it will sometimes give you a shorter one than the document supports. Whether that tendency worsens as the list grows is an empirical question with a measurable answer, and it has not been measured in the setting where it matters most for production systems: structured extraction under a schema.

Two literatures bracket the question without answering it. Long-context work measures how retrieval degrades with input length and with the position of the evidence, and multi-needle variants report that retrieval falls as the number of inserted facts rises [1]. Structured-output work measures the cost of constrained decoding on validity and on accuracy, and reports real losses [2]. Neither varies the number of items to emit while holding input length fixed, and neither compares decoding regimes on that axis.

The distinction matters because the two candidate explanations make opposite predictions. If degradation is a property of the model’s attention over a long list of targets, a grammar

should not help. If it is a property of the model choosing when to stop, a grammar that rejects incomplete output should help a great deal. The data support the second.

Contributions. We fix k , the number of items to emit, as an evaluation axis and hold input length constant by constructing documents rather than annotating them, which makes k exact and the gold set free. The result is reported as a slope, not an existence claim: the effect is present under unconstrained decoding and absent under a schema, inverting the expectation that grammars cost accuracy. A constraint that any study on this axis inherits is made explicit. Count, density and length are mechanically linked, so fixing two determines the third and decides the direction of the residual bias.

2 Constructing documents with known k

Annotation is the obvious approach and the wrong one. If k is a property of whichever documents happened to be sampled and a human must recover it, k is measured with error that correlates with the outcome, since an annotator’s attention flags on exactly the long enumerations where the effect should live.

We construct instead. The item bank is drawn from SEC 10-K filings: sentences each naming exactly one organisation, each verified to appear verbatim in its source document. Filler is drawn from the same corpus with every sentence containing a capitalised run removed, so it cannot carry a legitimate item. A document is k item sentences injected at uniformly random positions into filler, padded to a constant 9,000 characters. Gold is the injected set.

Three properties follow: k is exact, input length is constant by construction rather than merely uncorrelated, and position is averaged over rather than confounded with k .

The item bank is screened by measurement, not by rule. Three syntactic filters for “which sentences transplant into a synthetic document” were tried and all three failed, leaving recall at $k = 1$ far below any usable baseline. The bank is therefore defined empirically: a candidate sentence is retained only if the extractor recovers it at $k = 1$ in a short document. 143 of 240 candidates survive.

A validity gate. Before any slope is fitted, recall at $k = 1$ must clear 0.80 for a cell to be interpretable: one item, one document, a sentence the extractor itself produced in the wild. This gate failed three times during construction, and each failure was informative. The decisive diagnosis came from holding the item and k fixed and varying only the filler:

filler	recall at $k = 1$
1,500 characters	0.667
4,000 characters	0.583
14,000 characters	0.417

Dilution alone is worth 0.25 recall. The filler had been selected *for containing no entity names*, which made it far more inert than any real prompt, where a passage selector retains only relationship-dense text. Document length was reduced accordingly.

3 The count–density–length trilemma

The dilution measurement exposes a constraint that binds every study of this axis. Item density is mechanically determined by the other two quantities: $\text{density} = (k \times \text{span length}) / \text{document length}$. Fixing any two fixes the third.

Held fixed	Varies	Residual confound
length and count (this work)	density	density aids high k ; effect is a lower bound
density and count	length	length suppresses high k ; an upper bound
length and density	—	k cannot vary

We hold length fixed, so item density rises with k : a $k = 16$ document is denser in items than a $k = 1$ document of the same length. Since dilution demonstrably suppresses recall, high- k documents receive a density advantage that works against the hypothesis. Our estimate is therefore conservative.

4 Design and pre-registration

The hypothesis, estimator and rejection rule were fixed before any cell was run. Rejection requires *both* a recall drop of at least 0.15 from $k \leq 3$ to $k \geq 16$ *and* $p < 0.05$ on the rank test. Bucket means are for display; estimation is per document.

Both arms use `qwen3:14b` at temperature 0 with a 32,768-token context and a 24,000-token generation ceiling, on 75 matched documents each at $k \in \{1, 4, 16\}$, 25 per level. The unconstrained arm is prompted for JSON and parsed post hoc, with a deliberately tolerant parser: a parse failure would score as a recall failure and confound the arm under test.

The ceiling control matters and is easy to get wrong. An earlier attempt used a 30B mixture-of-experts model, which writes 7,000 to 12,000 tokens unconstrained against an 8,192-token window. Four of six $k = 16$ documents truncated mid-array, parsed to zero items, and would have reported constrained decoding as hugely superior at the level that decides the study. That arm was discarded rather than repaired. In the reported runs, zero documents truncate and the maximum generation is 686 tokens.

5 Results

k	constrained	unconstrained	paired difference	p
1	0.440	0.640	+0.200	0.096
4	0.640	0.530	−0.110	0.110
16	0.554	0.426	−0.128	0.005

Table 1: Recall by item count and decoding regime, paired on identical documents.

The pre-registered test rejects for one arm and not the other. Unconstrained decoding drops 0.214 from $k \leq 3$ to $k \geq 16$ with Spearman -0.293 , $p = 0.011$, satisfying both conditions.

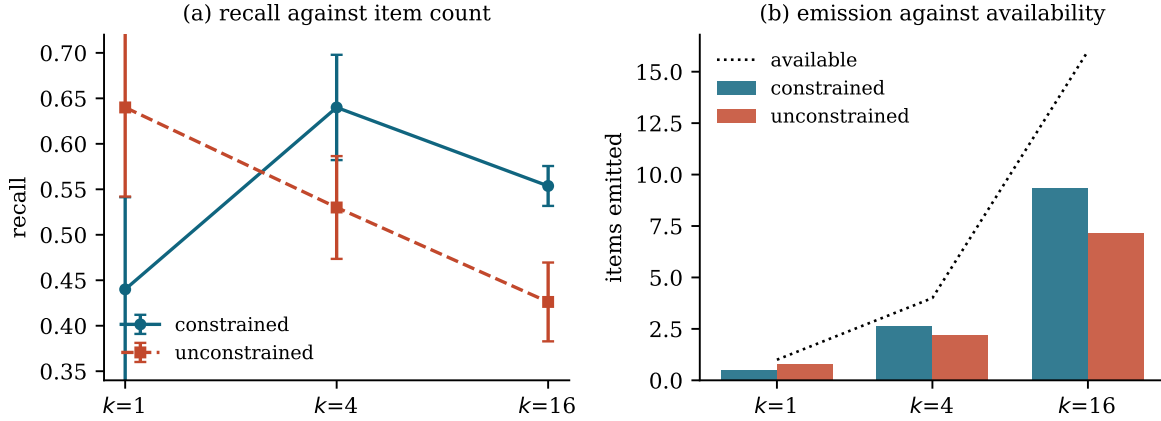


Figure 1: (a) Recall against item count for both decoding regimes, 75 matched documents per arm, error bars one standard error. The arms cross between $k = 1$ and $k = 4$. (b) Items emitted against items available. The constrained arm tracks availability more closely as k grows; the unconstrained arm falls away.

Constrained decoding drops -0.114 , meaning recall rises, with Spearman $+0.041$, $p = 0.73$, satisfying neither.

The $k = 1$ comparison favours unconstrained decoding by 0.200 at $p = 0.096$. Per-document recall at $k = 1$ is close to a Bernoulli draw, and the variance arithmetic implies 106 documents are needed to detect a 0.15 difference at 80% power against the 25 run. We report it and do not interpret it.

Mechanism. At $k = 16$ the constrained arm emits 9.4 of 16 available items and the unconstrained arm 7.2, at precision 0.946 and 0.948 respectively. The difference is not a precision–recall trade: the unconstrained arm simply produces fewer items, and produces relatively fewer as the list lengthens. A schema makes an incomplete array unparseable, so generation continues past the point where the model would otherwise stop.

Cost. Unconstrained decoding is not cheaper. It costs $1.1\times$ the tokens at $k = 1$ and $0.7\times$ at $k = 16$: on this model it is both less accurate at high k and no less expensive.

6 Scope conditions

One model, dense, one family; nothing here separates architecture from decoding regime. The reasoning-mode flag on this stack changes only which response field the same tokens arrive in, verified by identical evaluation counts, so this work says nothing about inference-time reasoning. The item bank is drawn from a single domain and screened by one extractor, so absolute recall levels are not comparable across corpora. The slope is the estimand. The $k = 1$ arm is underpowered by the study’s own arithmetic.

7 Conclusion

On the axis of how many items a model must emit, the grammar is not what costs recall. Schema-constrained decoding holds recall flat across k , and removing it produces a decline that clears a pre-registered threshold. The mechanism is a stopping decision rather than a retrieval failure, which is why removing the decision fixes it.

For practitioners the implication is direct: free-form generation with post-hoc parsing, often proposed as an escape from grammar masking, trades away recall on the long-list cases where extraction is most valuable, and saves nothing. For evaluation, a single F_1 over a benchmark whose k distribution is unstated is not comparable across benchmarks, because it averages over a mix nobody reported.

Reproduction. Code, pre-registration, the constructed evaluation set and all 450 raw model responses are released at <https://github.com/quantraunak/cardinality-eval>. The reported result re-scores without a GPU.

References

- [1] LangChain. Multi needle in a haystack. <https://www.langchain.com/blog/multi-needle-in-a-haystack>, 2024.
- [2] S. Geng et al. JSONSchemaBench: A rigorous benchmark of structured outputs for language models. arXiv:2501.10868, 2025.